First Assessment 2079

Subject: **Advanced Data Mining** — *Full Marks:* 45

Course No: **MDS 602** — *Pass Marks:* 22.5

Level: **MDS /II Year /III Semester** — *Time:* 2hrs

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A [3 ×5 = 15]

1. What is data mining? Describe about Major issues in Data mining?
2. Write down the steps to find the principal component of a data set.
3. What is data warehouse? Compare OLTP and OLAP.
4. What are categorical data? How can you handle the categorical data in association mining?
5. Explain Rule based classifier. How it is different than decision tree classifier.

Group B [5 × 6 = 30]

6. Classify the following attributes as binary, discrete, or continuous. Also classify them as qualitative (nominal or ordinal) or quantitative (interval or ratio). Some cases may have more than one interpretation, so briefly indicate your reasoning if you think there may be some ambiguity (Example: Age in years. Answer: Discrete, quantitative, ratio).
   a) Brightness as measured by a light meter.
   b) Angles as measured in degrees between 0 ∘ and 360 ∘.
   c) Bronze, Silver, and gold medals as awarded at the Olympics.
   d) Number of patients in a hospital.
   e) Ability to pass light in terms of the following values: opaque, translucent, transparent.
   f) Military rank.
7. Given the following data (in increasing order) for the attribute age: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. (a) Use smoothing by bin means to smooth the above data, using a bin depth of 3. (b) How might you determine outliers in the data?
8. Consider the data set shown in Table

Transaction ID	Items Bought
1	{a, d, e}
2	{a, b, c, e}
3	{a, b, d, e}
4	{b, c, e}
5	{b, d, e}
6	{c, d}
7	{a, b, c}
8	{a, d, e}
9	{a, b, e}

Compute the support for itemsets {e}, {b, d}, and {b, d, e} by treating each transaction ID as a market basket.

Use the results in part (a) to compute the confidence for the association rules {b, d} ⟶ {e} and {e} ⟶ {b, d}. Is confidence a symmetric measure?

Repeat part (a) by treating each customer ID as a market basket. Each item should be treated as a binary variable (1 if an item appears in at least one transaction bought by the customer, and 0 otherwise.)

OR

Construct the FP tree from the above transactional data table with minimum support of 33%.

9. Explain ID3 Algorithm with an example.
10. Describe Confusion Matrix with example. Define Accuracy, Precision, TPR, TNR, FPR, FNR of the classifier model.

OR

Given the following confusion matrix, determine Accuracy, Precision, TPR, TNR, FPR, FNR of the classifier model.

N=165	Predicted: No	Predicted: Yes	
Actual: No	50 TN	10 FP	60
Actual: Yes	5 FN	100 TP	105
	55	110	

$TPR = \frac{TP}{TP+FN}$

$TNR = \frac{TN}{TN+FP}$

$FPR =$

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Second assessment 2079

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Level: **MDS/II Year/III Semester**

Full Marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answer in their own words as far as practicable.

Attempt ALL questions.

Group A [5×3=15]

1. What is a super vector machine (SVM)? Why does SVM work fast and well?
2. Why does ensemble method classifier give better result? Explain main types of ensemble method.
3. How is the cluster quality measured? What are the factors affecting cluster quality. Explain with an example.
4. What is anomaly? Why it is important? Explain different types of outliers that occur in a dataset.
5. Explain distance-based outlier detection approach.

Group B [5×6=30]

6. Consider the 5 datapoints shown below: P1:(1,2,3), P2:(0,1,2), P3:(3,0,5), P4:(4,1,3), *P5:(5,0,1)*. Apply the K-means clustering algorithm, to group those data points into 2 clusters, using the Manhattan distance. Also calculate the SSE (Sum of Square Error). Suppose that the initial centroids are C1:(1,0,0) and C2:(0,1,1).

OR

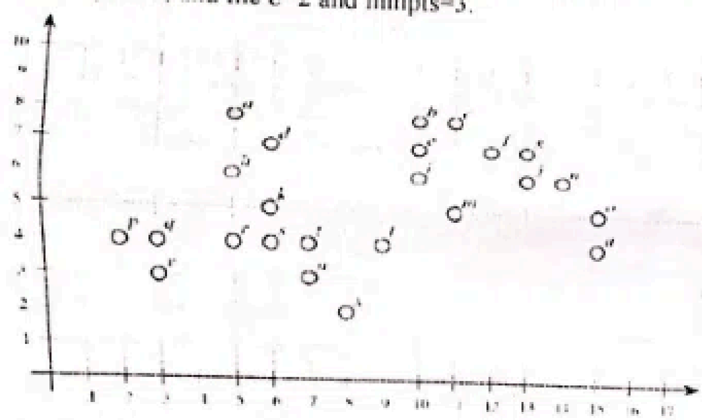
Cluster the following data and cluster the data using fuzzy C-means clustering algorithms. Assume k=2 and P=2.

X1	1	2
X2	2	3
X3	9	4
X4	10	1

7. Assume that database D is given by the table below. Follow single link technique to find clusters in D. Use any distance measure method.

	A	B	C	D
A	0	1	4	5
B		0	2	6
C			0	3
D				0

8. Consider the figure below and answer the following questions, we use the Euclidean distance between points, and the ε=2 and minpts=3.



a) List all the core points.
b) Is **a** directly reachable from **d**?
c) Is **o** density reachable from **i**? Show the intermediate points on the chain or the point where the chain breaks.

9. Explain different statistical based clustering method.

10. What are the approaches for density-based outlier detection. Explain Local Outlier Factor approach for outlier detection.

OR

A (0,0), B (1,0), C (1,1) and D (0,3) and K=2. Use LOF to detect one outlier among these 4 points. Use Manhattan distance method to measure the distance between the points.

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Master Level / Second Year /Third Semester/ Science
Data Science (MDS 602)
(Advanced Data Mining)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours

Candidates are required to give their answers in their own words as for as practicable.

Attempt All Questions

Group A [5 × 3 = 15]

1. What is data mining? Explain about data preprocessing.
2. What is sequential pattern mining? How Apriori is different than FP-Tree algorithm.
3. Explain different method for estimating a classifier's accuracy.
4. Explain K-means clustering algorithm.
5. What are the contextual and collective outliers in attribute of credit card company (name, age, job, address, annual-income, annual-expense, average-balance, credit-limit) and why?

Group B (5×6=30)

6. In real-world data, tuples with missing values for some attributes are a common occurrence. Describe various methods for handling this problem.
7. What is a rule-based classifier? Briefly discuss different types of rule-based classifier.

OR

The following table consists of training data from an employee database. The data have been generalized. For example, "31 . . . 35" for age represents the age range of 31 to 35. For a given row entry, count represents the number of data tuples having the values for department, status, age, and salary given in that row.

Department	Status	Age	Salary	Count
sales	senior	31-35	46k-50k	30
sales	junior	26-30	26k-30k	40
sales	junior	31-35	31k-35k	40
systems	junior	21-25	46k-50k	20

systems	senior	31-35	66k-70k	5
systems	junior	26-30	46k-50k	3
systems	senior	41-45	66k-70k	3
marketing	senior	36-40	46k-50k	10
marketing	junior	31-35	41k-45k	4
secretary	senior	46-50	36k-40k	4
secretary	junior	26-30	26k-30k	6

Given a data tuple having the values "systems," "26-30," and "46K-50K" for the attributes department, age, and salary, respectively, what would a naive Bayesian classification of the status for the tupie be?

8. A database has 5 transactions as below. Let min sup = 60% and min conf = 80%

TID	items bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y }
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I ,E}

Find complete set of frequent itemsets using Apriori algorithm. Also find the top two strong association rules between the items sets.

9. Assume that database D is given below. Follow complete link technique to find clusters in D. Also show the dendrogram.

	A	B	C	D
A	0	1	4	5
B		0	2	6
C			0	3
D				0

OR

Write an algorithm for DBSCAN and prove that in DBSCAN, for a fixed MinPts value and two neighborhood thresholds $\epsilon_1 < \epsilon_2$, a cluster C with respect to ϵ_1 and MinPts must be a subset of a cluster C′ with respect to ϵ_2 and MinPts.

10. Explain proximity-based outlier detection approaches.

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First Assessment 2080

Subject: **Advanced Data Mining**
Course No: **MDS 602**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. What is warehousing? What are the sources of data for building data warehouse.
2. List the differences between OLAP and OLTP.
3. List out the different techniques that can be applied for data mining with use case.
4. Why data pre-processing is important for data mining? List the possible task that can be done in data pre-processing phase.
5. Write different steps of K-NN algorithm.

Group B [5x6=30]

6. How can we build the Association based recommendation system. Explain with example.

OR

What is Apriori Algorithm. Discuss with its use case.

7. Explain data visualization techniques.

OR

Write ID3 Algorithm. With example discuss the pros and cons of the decision tree based classification technique.

8. Explain KDD with workflow.
9. What is Measurement of similarity? Why this is important in Data mining?
10. How ANN works? Explain the working of the perception for classification task.

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First Assessment 2081

Subject: **Advanced Data Mining**
Course No: **MDS 602**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5 × 3 = 15]

1. What are the different types of the warehouse? Explain with their uses.
2. Define KDD. What are the steps involved with KDD workflow?
3. What is association analysis?
4. Define data scaling and normalization with examples.
5. Write a K-NN algorithm with an example.

Group B [5 × 6 = 30]

6. Explain how classification can be used for data mining? Explain with examples.

OR

What is the Apriori Algorithm? Discuss with its use case.

7. List and explain the uses of the different types of data visualization techniques

OR

Write ID3 Algorithm. With an example, discuss the pros and cons of the decision tree based classification technique.

8. What is data pre-processing? List the data pre-processing techniques useful for numeric data
9. In the context of data mining, how should a data scientist decide whether to use supervised or unsupervised learning for a given problem? Discuss the key factors that influence this decision and provide examples of scenarios where each type of learning would be appropriate.
10. How does ANN work? Explain the working of the perceptron neuron for classification tasks.

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Second Assessment 2081

Subject: **Advanced Data Mining**
Course No: **MDS 602**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. What is warehousing? What are the sources of data for building a data warehouse?
2. What limitations of OLTP are solved in OLAP so that it can be used for data mining?
3. List out the different techniques that can be applied for data mining with use cases.
4. What is KDD? List the possible tasks that can be done in the data pre-processing phase.
5. Write different steps of the K-NN algorithm.

Group B [5x6=30]

6. What is Clustering? Explain giving DB Scan as an example algorithm.

OR

What is the Apriori Algorithm? Discuss with its use case.

7. Explain data visualization techniques.

OR

Explain the working of SVM with examples.

8. What is Ensemble techniques? Explain with example.
9. What is an outlier? How can clustering be used for outlier detection?
10. How can we build the Association based recommendation system? Explain with examples.

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2081

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Master Level / Second Year /Third Semester/ Science
Data Science (MDS 602)
(Advanced Data Mining)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours

Candidates are required to give their answers in their own words as for as practicable.

Attempt All Questions

Group A [5 × 3 = 15]

1. List the differences between OLAP and OLTP.
2. What is data pre-processing? List the data pre-processing techniques useful for numeric data.
3. Explain the senario where Ensemble methods are very useful.
4. What is association analysis?
5. Explain Fuzzy Clustering technique.

Group B (5×6=30)

6. What is the kernel function in SVM? Explain Gaussian Kernel Radial Basis Function (RBF) with examples.

OR

What is the frequent pattern growth approach for Association analysis? Discuss how this solves the issue of Apriori algorithm.

7. Explain different types of data visualization techniques.

OR

Explain the evaluation techniques that can be used for evaluating clustering models.

8. Explain the concept of Knowledge Discovery in Databases (KDD) and its importance in data mining. Discuss the key steps involved in the KDD process and provide examples to illustrate each step.
9. What is a decision tree? How can we construct a decision tree based classification model? Explain with appropriate examples.
10. A telecommunications company, Telecom Co, wants to identify unsual call patterns within its large dataset of customer call records. They have decided to use clustering techniques for outlier detection. Define what an outlier is in this context and explain how TelecomCo can use a clustering based approach to detect these outliers.

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First Assessment 2082

Subject: **Advanced Data Mining**
Course No: **MDS 602**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A **[5x3=15]**

1. What are the different types of the warehouse? Explain with their uses. enterprise / operational
2. What limitations of OLTP are solved in OLAP so that it can be used for data mining?
3. With example differentiate the uses of the clustering and classification.
4. Define data scaling and normalization with example.
5. Write K-NN algorithm.

Group B **[5x6=30]**

6. Explain how classification can be used for data mining? Explain with examples.

OR

What is Apriori Algorithm. Discuss with its use case.

7. List and explain the uses of the different types of data visualization techniques.

OR

Write ID3 Algorithm. With example discuss the pros and cons of the decision tree based classification technique.

8. Define KDD. What are the steps involved with KDD workflow?
9. Define Bayes classifier. How can it be used for classification task?
10. How ANN works? Explain the working of the perception for classification task.

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SCHOOL OF MATHEMATICAL SCIENCES

Second Assessment 2082

Subject: **Advanced Data Mining**
Course No: **MDS 603**
Level: **MDS / II Year / III Semester**

Full marks: **45**
PassMarks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. How to build a data warehouse? Explain in the context of retail business?
2. Explain the concept of "slice" and "dice" in OLAP. Provide an example scenario to illustrate how these operations can be used for data analysis.
3. Discuss the descriptive data mining techniques.
4. What are the differences between Normalization vs Standardization in numerical data?
5. List out the advantages and disadvantages of the K-NN algorithm.

Group B [5x6=30]

6. What is Clustering? Differentiate between K-Means and DBScan algorithm.

OR

What is classification? Discuss any decision tree induction technique with the help of an example.

7. Explain exploratory data analysis techniques discussing the context of data mining.

OR

What is the kernel function in SVM? Explain Gaussian Kernel Radial Basis Function (RBF) with examples.

8. Suppose you have to solve the multi-class classification problem. After research you concluded that none of the single ML models are capable of classifying all labels correctly. Explain how you deal with this situation?
9. A financial institution, ABC Bank, wants to identify fraudulent transactions within its vast dataset of customer transactions from the core banking system. They decide to use clustering techniques for outlier detection. Explain what an outlier is in this context and describe how ABC Bank can use clustering to detect these outliers, providing examples to illustrate the process.
10. Company XYZ, an online retail platform, wants to improve its product recommendation system to increase sales and enhance customer satisfaction. They are considering implementing an association-based recommendation system. Describe how Company XYZ can build this system, explaining the steps involved and providing examples of how the recommendations would work in practice.

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